Food Redistribution Management Database

Database design and testing in a 12-week team project

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A database for coordinating surplus food

The problem

Restaurants, pantries and nonprofits needed a shared record of available food, destinations and donation status.

My contribution

Contributed to relational database design and testing, with attention to data integrity and dependable information retrieval.

The team solution

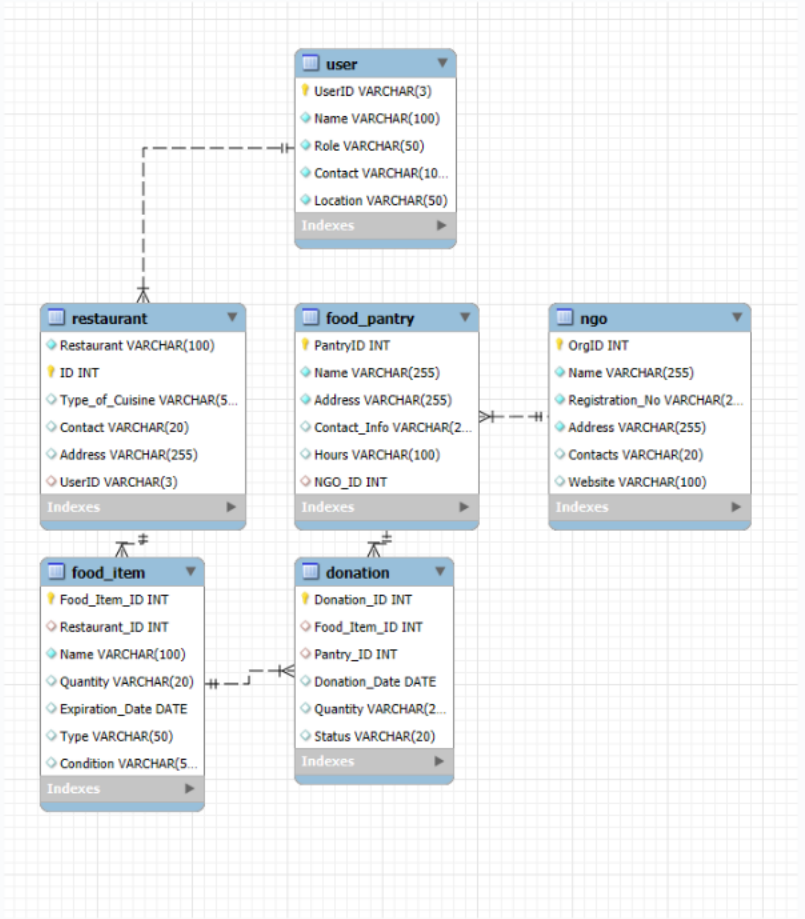
Linked users, restaurants, food items, pantries, NGOs and donations in a MySQL database.

Why it mattered

Made the flow of a donation traceable within the prototype, supporting inventory and donation reporting.

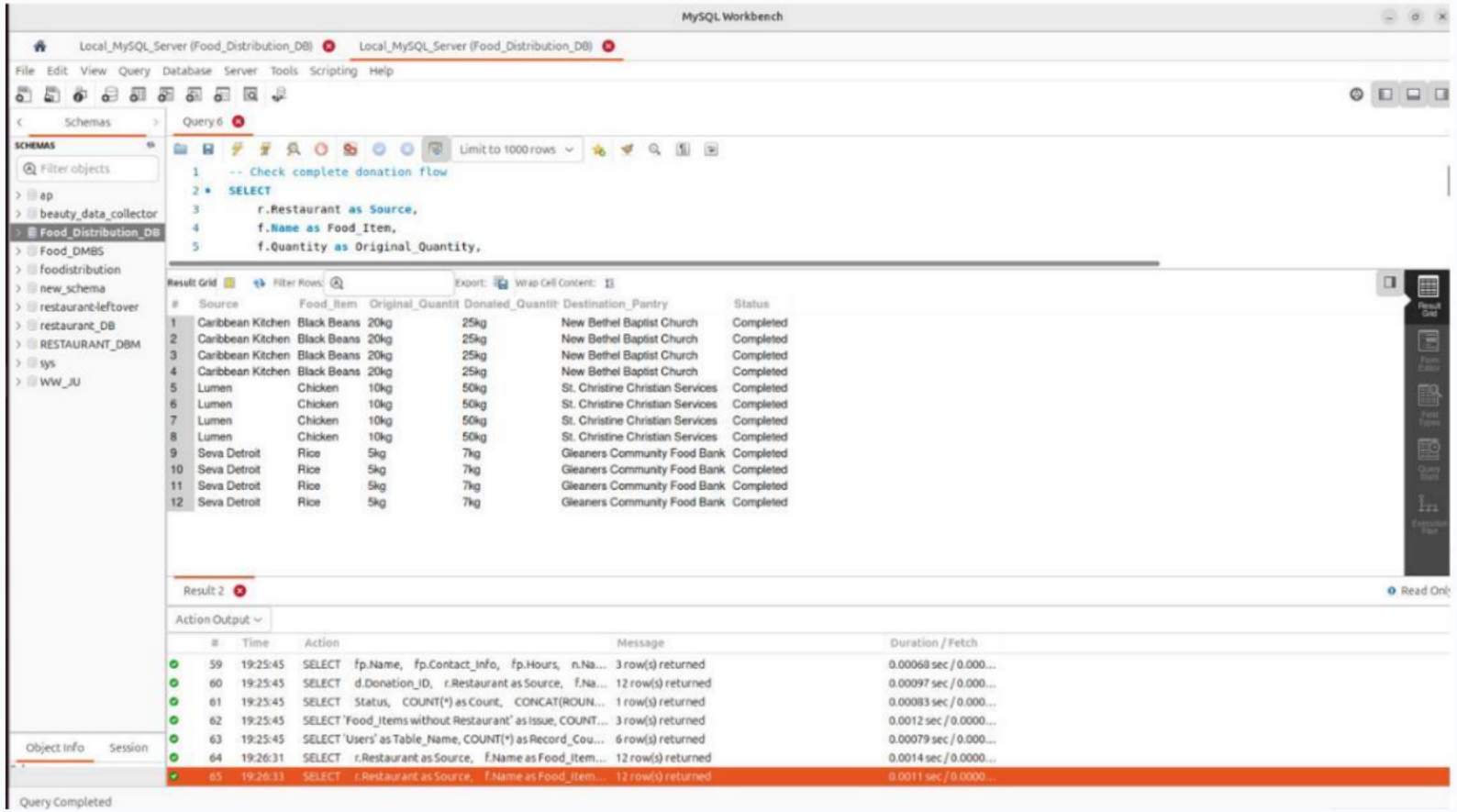
Six related entities connected the workflow

Restaurant to food item
Food item to donation
Donation to food pantry
Food pantry to NGO



Primary and foreign keys connected records across the model. Normalization reduced repeated information.

SQL made donation records traceable



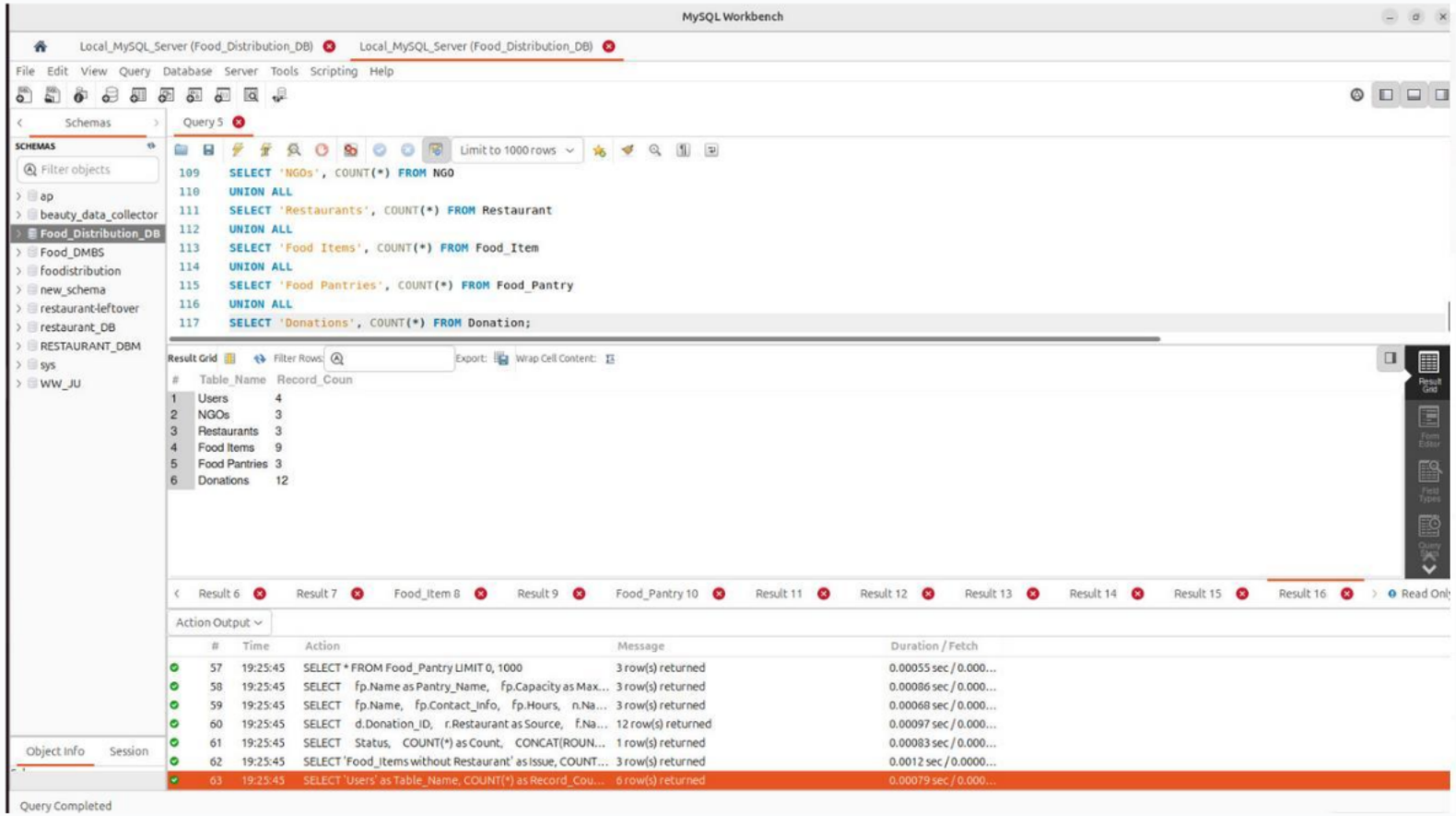
A connected view

The query connected the source restaurant, food item, donated quantity, destination and status.

Analyst relevance

Joins and clear entity relationships turned separate tables into information users could review.

Testing checked the data behind the workflow



Database checks

Reviewed table contents, keys and relationships to identify inconsistent or invalid records.

Functional checks

Tested search filters, permissions and tracking behavior, and supported defect resolution.

The model supported operational reporting

Inventory availability

Identify food by type, quantity and expiration date to support allocation and timely distribution.

Donation tracking

Connect food sources and destinations to review donation status and follow up on pending items.

Donor and pantry activity

Summarize contributions and pantry activity for coordination and project reporting.

Data quality follow-up

Use exceptions and relationship checks to flag incomplete records before reporting.

What the project demonstrated

My contribution

Combined database design with validation and functional testing as part of a four-person academic team.

Skills demonstrated

Relational modeling, MySQL, MySQL Workbench, Lucidchart, data validation and collaborative problem solving.

Next improvement

Store quantities as numeric values with explicit units, and enforce available-stock and expiration rules.

Scope

The deliverable was an academic prototype. Community impact and production performance were not measured.